

Examining Behavioral Resurgence in Virtual Reality

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Figure 1: The image on the left displays the observation two-way mirror and the bucket of ping-pong balls from which the participants grab ping-pong balls to make responses. The image on the right displays the table holding the response baskets. In between the response baskets is the iPad on which the points are displayed.

ABSTRACT

Behavioral resurgence, the reappearance of a previously extinguished behavior, poses challenges in applied settings. Traditional human operant studies often rely on in-vivo procedures, but emerging technologies such as virtual reality (VR) offer immersive environments that may alter motivating operations and contextual control. This study compared resurgence patterns in VR and in-vivo contexts using a three-phase procedure: Baseline, Differential Reinforcement of Alternative Behavior (DRA), and Extinction. Thirty undergraduate participants completed two sessions, one in VR and one in-vivo, involving arbitrary responses (placing virtual or physical ping-pong balls into baskets) under identical contingencies. Results indicated that resurgence occurred in both contexts, with no statistically significant differences in target responding across phases. However, VR sessions demonstrated slightly higher extinction rates and unique interaction patterns, suggesting VR can replicate operant contingencies while offering opportunities for enhanced experimental control and gamification. These findings support VR as a viable modality for studying complex behavioral processes and highlight future directions for leveraging immersive technology in behavioral research.

Index Terms: Virtual Reality, Behavioral Resurgence, Experimental Analysis of Behavior

1 INTRODUCTION

Behavioral resurgence poses significant challenges in applied and experimental settings, particularly when interventions aim to reduce dangerous or maladaptive behaviors. Resurgence typically occurs when reinforcement conditions worsen for an alternative response after the target behavior has been placed on extinction [6]. This phenomenon is often studied using a three-phase procedure: Baseline (reinforcement of the target response), Differential Reinforcement of Alternative Behavior (DRA), and Extinction (no reinforcement for any response). While these procedures are well

established in basic and applied research, most studies have relied on in-vivo contexts or simple computer-based tasks [8].

Human operant research faces unique challenges, including schedule insensitivity and weak establishing operations [9]. For example, participants often fail to extinguish responding during the final phase, leading to undifferentiated patterns across target and alternative responses [7]. These limitations raise questions about whether traditional laboratory environments provide sufficient motivating operations for contrived reinforcers such as points or tokens.

Virtual reality (VR) offers a promising alternative by creating immersive, highly controlled environments that may enhance engagement and contingency sensitivity. VR has been widely adopted in fields such as surgical training, social skills instruction, and safety interventions [2, 4, 1]. Its ability to replicate real-world physics and provide dynamic feedback makes it an ideal platform for operant research. Prior work comparing motor tasks in VR and physical environments found similar performance outcomes, suggesting that VR can approximate real-world behavior under controlled conditions [3].

Despite these advantages, VR remains underutilized in the study of resurgence. Existing research has explored gamification in computer-based resurgence task [7], but few studies have examined whether immersive VR contexts influence resurgence patterns compared to traditional in-vivo procedures. The current study addresses this gap by evaluating whether VR-based resurgence procedures produce comparable behavioral patterns to in-vivo settings and whether VR offers unique benefits, such as increased extinction rates or differentiated responding during the extinction phase.

2 METHOD

Thirty undergraduate participants were recruited through an online research system and campus advertisements. Each participant completed two sessions, one in a virtual reality (VR) context and one in an in-vivo context, with the order counterbalanced using block randomization. Sessions were conducted in the same laboratory space to minimize environmental variability. Participants were screened for conditions such as motion sickness and migraines to ensure safe use of VR equipment.

Both contexts implemented a standard three-phase resurgence procedure: Baseline, DRA, and Extinction, with each phase lasting five minutes. In the VR condition, participants interacted with

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Figure 2: The image displays what a participant was able to see when using the VR program.

a Unity-based program via an Oculus headset and controllers, placing virtual ping-pong balls into baskets that replicated the physical setup to scale (Figure 2). The program incorporated real-world physics and automated point delivery. In the in-vivo condition, participants performed the same task using physical materials, with reinforcement delivered manually through a visible point counter (Figure ??). Responses were recorded in one-minute bins, and resurgence was defined as any increase in target responding during Extinction compared to the final two minutes of DRA.

3 RESULTS

Behavioral responding varied systematically across phases, confirming the expected resurgence pattern in both VR and in-vivo contexts. Out of 60 data sets, 76.7% exhibited resurgence, defined as an increase in target responding during Extinction compared to the final two minutes of DRA. This effect was consistent across participants regardless of session order, indicating that the contingency shift reliably evoked previously extinguished behavior.

Although group-level analysis revealed no significant differences between VR and in-vivo conditions ($p = .417$), the phase effect was highly significant ($p < .001$), underscoring the influence of programmed contingencies. Mean target responding was highest during Baseline, decreased sharply during DRA, and rebounded during Extinction, a classic resurgence profile. However, minute-by-minute analysis revealed subtle but noteworthy distinctions between contexts. Participants in the VR-first group tended to reallocate responding to the target earlier in Extinction and maintained higher target rates during later bins, whereas the in-vivo-first group showed more persistent alternative responding, suggesting that VR may alter response allocation dynamics even when overall resurgence magnitude remains similar.

Extinction patterns further differentiated the two contexts. Evidence of extinction—defined as zero responses for at least one minute—occurred in 63% of VR sessions compared to 31% of in-vivo sessions. This suggests that immersive environments may facilitate faster disengagement from previously reinforced behaviors, possibly due to heightened contextual salience or increased sensitivity to contingency changes. Complete elimination of responding was rare in both contexts, echoing prior findings of persistent responding in human operant studies [8, 5].

Visual analysis of individual data sets revealed variability in resurgence onset and magnitude. Some participants demonstrated immediate resurgence within the first minute of Extinction, while others exhibited gradual increases over time. Interestingly, participants who experienced VR second often showed more rapid extinction of alternative responses, potentially reflecting carryover effects from prior exposure to contingencies.

4 DISCUSSION

The findings demonstrate that virtual reality can effectively replicate the contingencies of a traditional resurgence procedure, producing patterns of responding comparable to those observed in in-vivo contexts. Although no significant differences emerged be-

tween conditions at the group level, the immersive VR environment offered unique advantages, including a greater frequency of extinction for target and alternative responses. This suggests that VR may enhance sensitivity to programmed contingencies, potentially due to increased engagement or contextual control.

The lack of systematic differences between VR and in-vivo conditions may be attributed to the high degree of similarity between the two environments. The VR program was intentionally designed to mirror the physical setting, which likely minimized contextual variability. While this approach ensured procedural consistency, it may have limited the potential for VR-specific effects. Additionally, short phase durations and the absence of control responses may have constrained the ability to differentiate resurgence from extinction-induced variability.

These results align with previous human operant research showing persistent responding during extinction [8, 5] and extend this literature by demonstrating that VR can serve as a viable platform for studying complex behavioral processes. Importantly, VR offers opportunities for experimental innovation, such as manipulating environmental features or incorporating game-like elements to alter motivating operations.

5 CONCLUSION AND FUTURE WORK

This exploratory study provides preliminary evidence that VR-based resurgence procedures can replicate behavioral patterns observed in traditional laboratory settings. While no significant differences emerged between conditions, VR offered practical advantages such as immersive engagement and automated control, suggesting its potential as a research and intervention tool. Future work should capitalize on these strengths by incorporating gamification elements to enhance motivation, designing more naturalistic VR environments for ecological validity, and extending session durations to better capture extinction dynamics. Additionally, integrating control responses and exploring clinical applications could advance both theoretical understanding and practical use of VR in behavior analysis.

REFERENCES

- [1] Ü. Çakıroğlu and S. Gökoğlu. Development of fire safety behavioral skills via virtual reality. *Computers & Education*, 133:56–68, 2019. 1
- [2] P. Cipresso, I. A. C. Giglioli, M. A. Raya, and G. Riva. The past, present, and future of virtual and augmented reality research: a network and cluster analysis of the literature. *Frontiers in psychology*, 9:2086, 2018. 1
- [3] E. Ebrahimi, S. V. Babu, C. C. Pagano, and S. Jörg. An empirical evaluation of visuo-haptic feedback on physical reaching behaviors during 3d interaction in real and immersive virtual environments. *ACM Transactions on Applied Perception (TAP)*, 13(4):1–21, 2016. 1
- [4] A. Froli, G. Savarese, F. Di Carmine, A. Bosco, E. Saviano, A. Rega, M. Carotenuto, and M. C. Ricci. Children on the autism spectrum and the use of virtual reality for supporting social skills. *Children*, 9(2):181, 2022. 1
- [5] A. Galizio, J. E. Friedel, and A. L. Odum. An investigation of resurgence of reinforced behavioral variability in humans. *Journal of the experimental analysis of behavior*, 114(3):381–393, 2020. 2
- [6] K. A. Lattal, C. R. Cançado, J. E. Cook, S. L. Kincaid, T. D. Nighbor, and A. C. Oliver. On defining resurgence. *Behavioural Processes*, 141:85–91, 2017. 1
- [7] C. N. Martinez-Perez, C. M. Ritchey, T. Kuroda, and C. A. Podlesnik. Blackouts can serve as a contextual feature and enhance resurgence. *Behavioural Processes*, 195:104587, 2022. 1
- [8] V. Saini, W. E. Sullivan, A. R. Craig, N. M. DeRosa, A. Rimal, K. Derrenbacher, and H. S. Roane. Responding fails to extinguish during human-laboratory experiments of resurgence. *The Psychological Record*, 71(3):325–336, 2021. 1, 2
- [9] E. Shimoff, B. A. Matthews, and A. C. Catania. Human operant performance: Sensitivity and pseudosensitivity to contingencies. *Journal of the experimental analysis of behavior*, 46(2):149–157, 1986. 1